

of Platform as a Service.

- **Infrastructure as a Service:** IaaS is a service model where the entire responsibility of managing the infrastructure, installing and configuring the runtime environment, and the configuration of high availability and fault tolerance is with the cloud consumer. Cloud consumers have no control over cloud resources (physical resources), but can configure or control operating systems, networks, storage, firewalls, and so on.

Cloud deployment models

Cloud deployment models specify where the cloud resources are available.

- **Public cloud:** Here, the cloud resources are made available to the public over the Internet. Since multiple organisations and individual users avail and share these resources, security is important while using the public cloud.
- **Private cloud:** Here, the cloud resources are made available to specific units within the perimeter of an organisation. Security and governance is maintained by the organisation, and resources are shared by multiple business units of the same organisation.
- **Hybrid cloud:** Organisations where security is the main concern use the hybrid cloud. Different types of cloud deployment models are used to address security concerns and for effective utilisation of resources.
- **Community cloud:** Here, the cloud resources are made available to a specific community of cloud consumers, such as life science based organisations or banks.

Essential characteristics

Essential characteristics identify the cloud based on the services that are made available by the provider.

- **On-demand self-service:** Services or resources are available on demand here, which is far better than a scenario where resources



Figure 2: DevOps — people, processes and tools

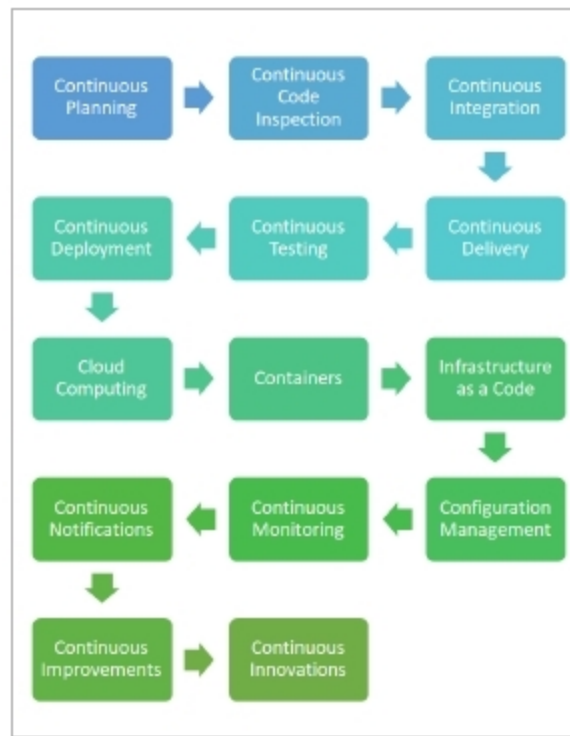


Figure 3: Continuous practices

are acquired across weeks. Cloud consumers only need to provide a credit card and use the resources. Free tiers are also available up to certain limits. There is no human interaction required, except for some governance related restrictions to manage cost.

- **Broad network access:** Cloud resources or capabilities are available over the Internet or intranet in this case.
- **Resource pooling:** Resource pooling is one of the most essential parts of cloud resources. Here, capabilities are presented in a unified manner. Consider a scenario where you have two physical machines, each with 50GB RAM, and you want to create a virtual machine with 51GB RAM. Resource pooling helps here. The two physical machines have a total of 100GB RAM, and cloud computing provides you the option

to create a virtual machine with 51GB RAM. Cloud consumers are unaware of the physical machines, but can use the resources of the virtual machine as per their needs.

- **Rapid elasticity:** Cloud resources can be provisioned and deprovisioned as per need. Scaling up and down of resources can be configured automatically, based on resource usage.
- **Measured services:** Cloud service providers offer facilities to monitor resources, measure the usage, and bill based on that usage. The 'pay as you use' feature is popular for various use cases where resources are required for a specific time only.

DevOps

DevOps is a combination of two words — development and operations. When security is also a part of this term, it is known as DevSecOps.

DevOps is a philosophy or a culture and not a technology. It is important to understand that multiple dimensions need to sync logically to create a roadmap for cultural transformation in an organisation.

People are extremely important in this triplet. They need to be made aware about the benefits of this cultural change; they need to realise why it is important to change and move away from traditional practices. Training and skills development are much needed to bring about this change.

Processes also play an important role in culture transformation.

They help to monitor and measure compliance. It is important to track the progress of all the aspects involved in the cultural or digital transformation.

DevOps has to have a tools agnostic approach. Any tool can be used to bring the cultural change. It can be open source or commercial. Tools don't matter, as all provide almost the same kind of features. It is all about the men and women behind the machines or tools, and not the tools.

DevOps includes multiple practices, as shown in Figure 3.